Sports Participation and Happiness

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Across cultures and generations, people have played sports for many reasons. Some for the spirit of competition, others for the entertainment of the game, and others still for simple exercise. Many individuals have participated in sports for the joy and contentment it brings them. The main focus of this paper lies in this connection. Is participating in a sport positively associated with one's happiness?

Further investigation can reveal potential moderators that may affect the strength or direction of this relationship. Gender plays a significant role in many aspects of society due to differing societal expectations and opportunities. How does gender affect this relationship? Additionally, individuals in different life stages experience the world in distinct ways. How does age affect this relationship?

The first hypothesis is that sports participation is positively associated with happiness. There are a couple possible reasons this may be true. First, the physical activity from participating in sports has positive effects on mental well-being. (White et al., 2024). The more one participates in a sport, the more physically active they are, leading to better mental well-being. Second, participating in sports with others allows individuals to build stronger relationships and gain social support. Social support is one of the predictors for happiness that the *World Happiness Report* uses, indicating its importance in happiness outcomes (Helliwell et al., 2017).

Additionally, if an individual participates in sports through clubs or organizations, it gives them the opportunity to find community. Engagement in organized sports may also build social capital, a theory popularized by Robert Putnam in *Bowling Alone*. Putnam (1995) defines social capital as "features of social organization such as networks, norms, and social trust that facilitate coordination and cooperation for mutual benefit." In his work, Putnam recognized that organizational membership provides a source of social connectedness, which can build a society's social capital, and in turn, its well-being.

With respect to gender as a moderator, the next hypothesis is that the relationship between sports participation and happiness is stronger for men than for women. Using social-role theory, which proposes that gender differences are due to inherited gender roles instead of simply biological differences, some explanations emerge (Clopton, 2012). Women are often seen as more communal than men, which may influence the types of benefits they derive from sports participation. Traditional masculine roles encourage competition and achievement, while traditional feminine roles encourage cooperation and social bonding. Consequently, in competitive sports environments, men may be more positively impacted as they align with their societal expectations. Historical disparities may also have impacts on women's access to sports. As a result, women may not be able to reap as many of the benefits of sports participation, as their opportunities can be limited.

With respect to age as a moderator, the final hypothesis is that the relationship between sports participation and happiness is stronger for younger people than older people. In recent years, adolescents and young adults have seen a significant decline in social engagement, in part due to increased digital interactions. This age group tends to spend more time with friends and others than other age groups, so the decline indicates a significant loss. Consequently, this age group is more sensitive to socialization experiences, such as sports participation, than others (Kannan et al., 2022). Thus, sports participation may have a more profound impact on younger people.

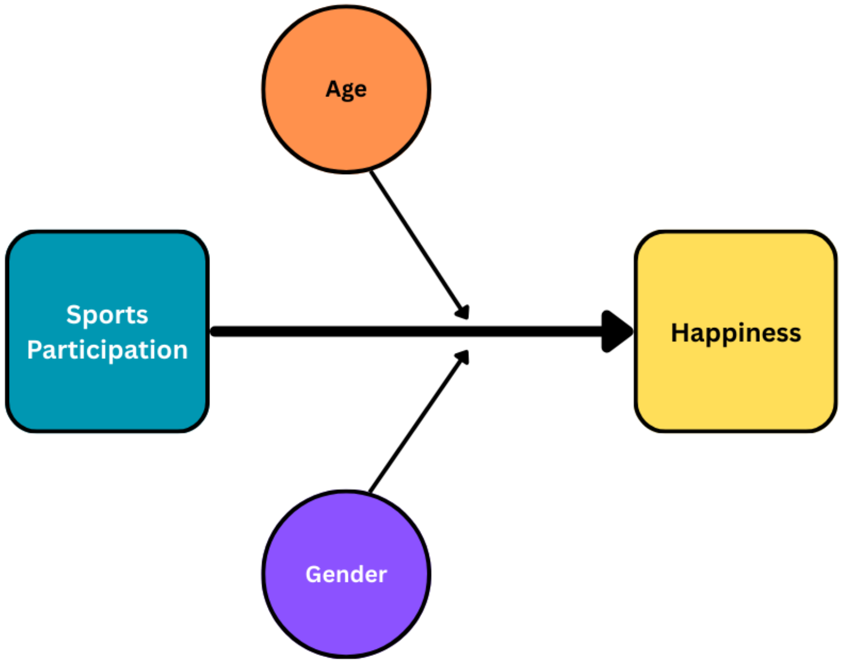

Figure 1: Moderator Diagram

The dataset used for this paper is from Wave 7 of the World Values Survey, specifically data collected in the United States in 2017. This survey is a global study of what people value in life. The sample for the United States portion was of 5,375 adults age 18 and older, with 2,596 interviews completed via phone and the web. To conduct the analysis, all "No Answer" responses to questions were removed. Less than 25 data points were removed, which is negligible given the overall sample size. This sizable sample allows for more accurate and generalizable conclusions.

The variable "Active/inactive membership: sports or recreational organization" is used to measure the level of sports participation. Participants could choose from three options: "active member," "inactive member," or "don't belong." Active members represent a higher level of sports participation, while those who don't belong represent a lower level, with inactive members in between at a moderate level. While this variable may not account for individuals who participate in sports outside of organizations, it still encompasses a broad range of involvement by capturing formal affiliations, which often entail more sustained and structured participation.

The variable “Satisfaction with your life” is used to measure the level of happiness. Participants were asked “All things considered, how satisfied are you with your life as a whole these days?” They could answer on a scale from 1 to 10, with 1 representing completely dissatisfied and 10 representing completely satisfied. Life satisfaction is a common measure of happiness used in prominent research such as the *World Happiness Report* (Helliwell et al., 2017).

The variable “Sex” is used to indicate gender. Participants were categorized as either male or female, with no additional options provided. Furthermore, the sex of the participant was observed instead of asked by the interviewer rather than self-reported. The strict binary may not fully represent all the individuals in the sample, and interviewer assumptions may not accurately reflect participants’ gender identity or sex. These limitations should be considered when interpreting gender-related findings.

The variable “Age” is used to indicate age. Participants were asked to state their age in years. For analysis, this variable was recoded into three groups in the dataset with ranges 18-29, 30-49, and 50 or more years.

Typically, correlation is shown using scatterplots. However, membership status in a sport or recreational organization is an ordinal variable, as it consists of three categories with a natural order: “don’t belong,” “inactive membership,” and “active membership.” Consequently, this measure of sports participation cannot be directly used for scatterplots.

To visualize this relationship with a scatterplot, country-level data must be used. Drawing from the full World Values Survey Wave 7 (2017-2022) dataset, the percentage of respondents in each country who report active membership in a sports organization and the average life satisfaction score for that country are displayed in Figure 2. Only countries with available data

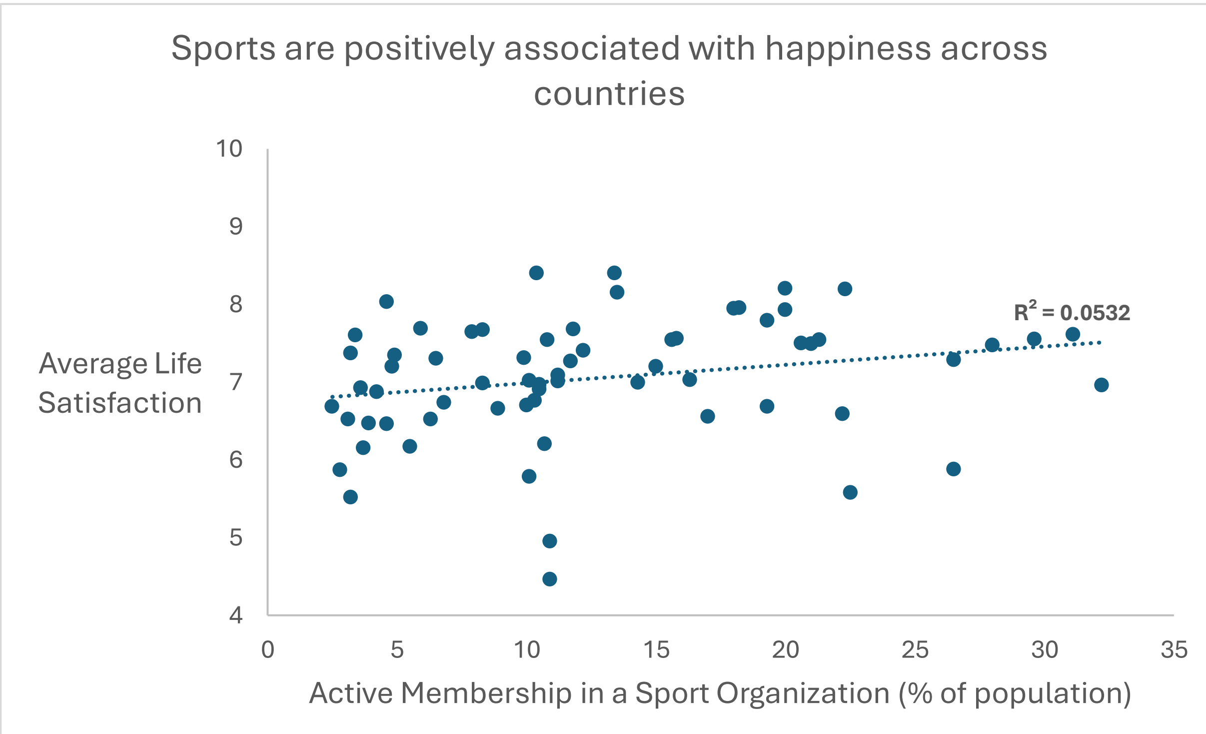

Figure 2

for both variables are included. Latvia was excluded from the plot due to an extreme outlier of over 90% active membership, which would have greatly skewed the plot. As a result, this analysis includes 66 total countries.

From the scatterplot, there appears to be a weak to moderate positive relationship between the percentage of respondents with active membership in a sports organization and the average life satisfaction in each country, with an R value of 0.231. Approximately 5% of the variance in average life satisfaction can be accounted for by the proportion of respondents who are active members.

To examine this relationship using the individual-level data from the United States, a side-by-side boxplot is presented in Figure 3.

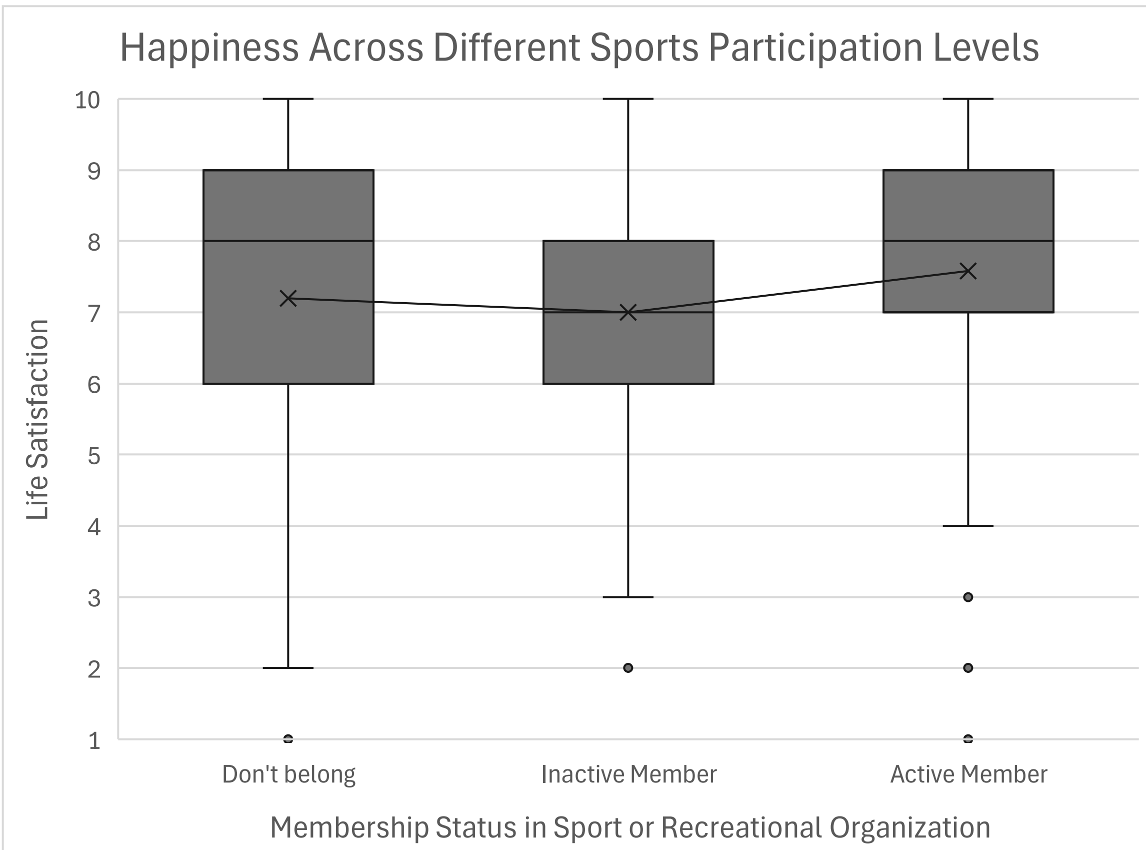

Figure 3

The spreads across the three membership statuses are relatively similar, although individuals who do not belong to sports organizations exhibit the largest spread. The interquartile range (middle 50%) of life satisfaction scores for active members lies between 7 and 9, slightly higher than inactive members (between 6 and 8) and non-members (between 6 and 9). Median life satisfaction scores fluctuate between 7 and 8 across all groups. On average, active members report slightly higher life satisfaction than the other two groups. The effect size between active members and non-members is 0.143, indicating a small difference. This weak evidence suggests the existence of a positive relationship between sports participation and happiness is not clearly established. However, it is important to note that while life satisfaction is measured on a 10-point scale, even small differences may still be meaningful, as most of the responses fall between 6 and 9. Overall, these findings offer limited but partial support for the the main hypothesis.

To examine gender as a moderating variable, separate side-by-side boxplots were produced for men (Figure 4) and women (Figure 5).

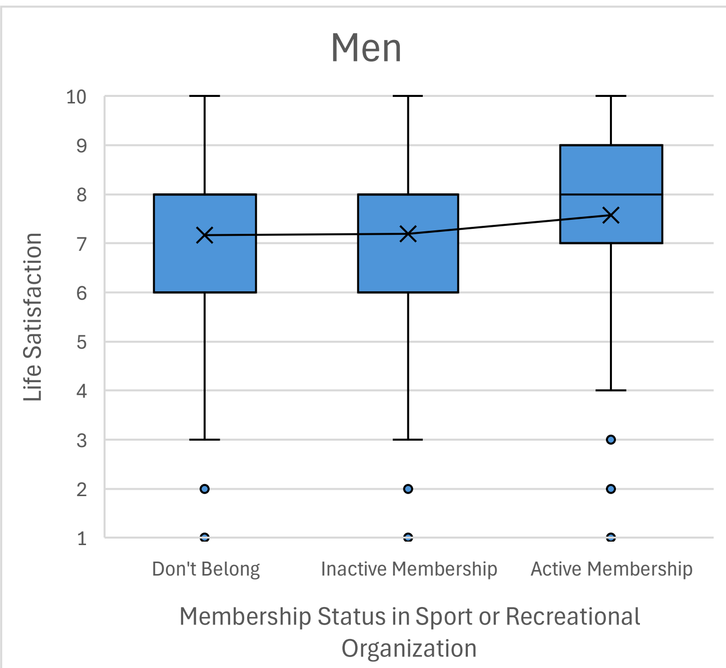

Figure 4

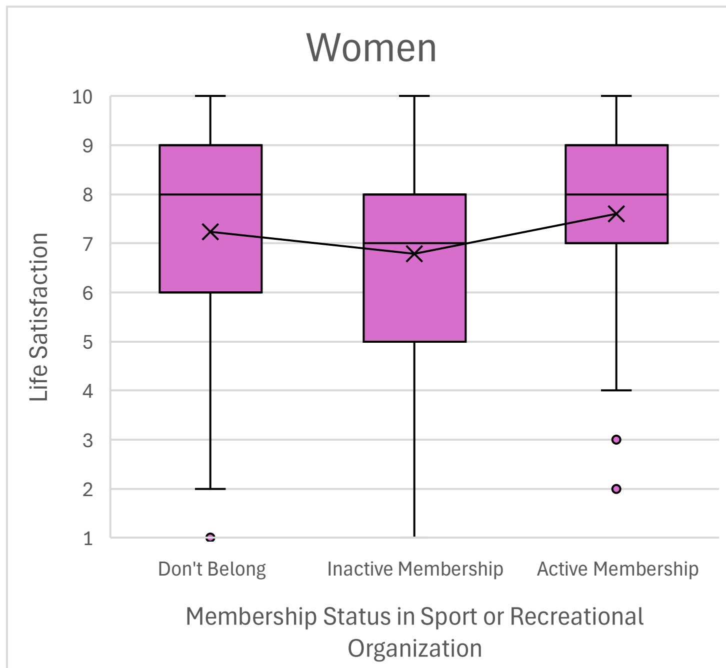

Figure 5

These plots reveal some differences in the relationship between sports participation and happiness by gender. Among men, life satisfaction appears to slightly increase with higher levels of sports participation. Among women, the trend is less consistent, with some fluctuation in life satisfaction across membership levels. Thus, the relationship appears stronger for men than for women, as predicted. However, as with Figure 3, there are limitations on analysis and interpretation due to the ordinal nature of sports participation level. If possible, side-by-side scatterplots would be more informative.

To explore age as a moderating factor, three side-by-side boxplot charts were created for each age group: 18-29, 30-49, and 50 or older, shown by Figures 6, 7, and 8, respectively.

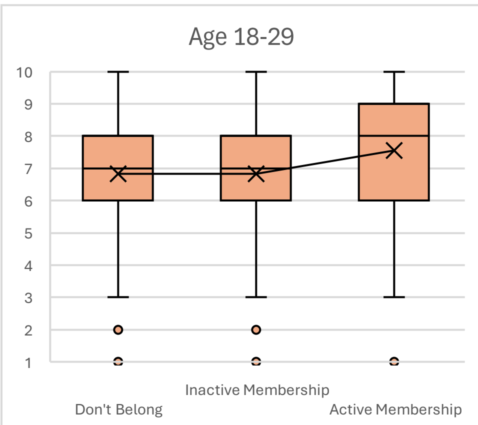

Figure 6

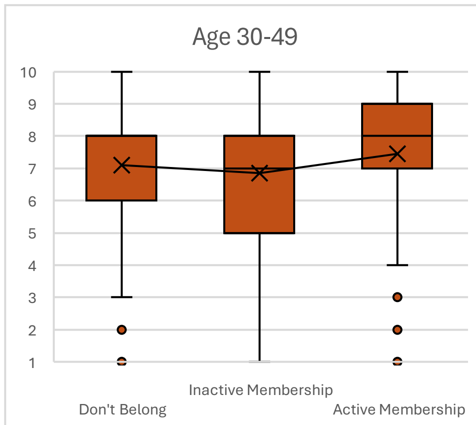

Figure 6

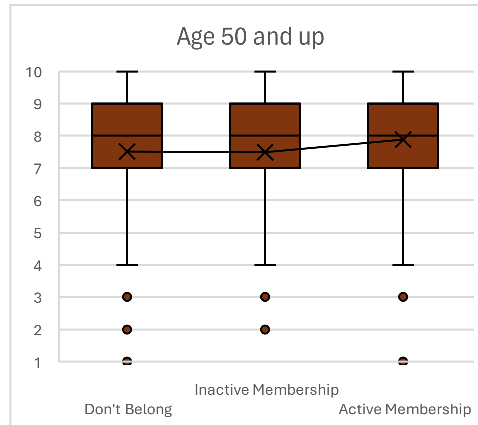

Figure 8

These plots show very similar mean life satisfaction trends across all age groups as the level of sports participation increases. This suggests that age is not likely a moderating factor, contradicting the hypothesis. One possible explanation is that the mental wellness benefits attained from physical activity, such as playing a sport, do not depend on age. Alternatively, adolescents and young adults may not be more sensitive to socialization experiences than other age groups, as suggested previously. People of all ages can benefit from social engagement and connection.

However, differences in median life satisfaction scores across age groups offer additional complexity. The younger two age groups follow identical paths, with equal medians at lower levels of participation and a small increase at the highest level. The oldest age group shows no change in median life satisfaction with increased level of sports participation. This measure may indicate that younger groups have a stronger positive relationship between sports participation and happiness. As with the previous figures, this analysis is quite blunt, so it is difficult to draw precise conclusions. More detailed patterns may emerge within these three age ranges if age was divided into smaller intervals.

The results from the data visualizations provide mixed of support for the three hypotheses. The nature of the data limited both the accuracy and depth of the analysis regarding the correlation between sports participation and happiness, as well as the role of potential moderators. In future research, it would be useful to measure sports participation based on the frequency of sports participation rather than sports organization membership alone. Such a measure would offer greater construct validity as well.

The reservations of using membership status in a sports organization may point to an alternative explanation for its positive relationship with happiness. It is possible that general organizational membership shares a similar association with life satisfaction. This would imply that the underlying mechanisms could be social support and social capital. Conducting similar analyses using membership statuses in other types of organizations could provide valuable insight into this possibility.

Future studies could also expand on the country-level analysis introduced in Figure 2. Prior research using the World Values Survey Wave 6 (2010-2014) found a positive association between sports and happiness (Balish et al., 2016). Further exploration of this relationship using more recent data could provide more support and uncover possible cultural and regional influences that may affect sports participation and happiness.

Additional research has examined differences in well-being between athletes who participate in team sports versus individual sports. For example, Pluhar et. al (2019) found that team sport athletes were less likely to experience anxiety or depression than individual sport athletes. This finding stresses the importance of social interaction and connection in mediating the relationship between sports participation and happiness.

Another intriguing area of research involves college students. How do college athletes compare in happiness and well-being to their nonathlete peers? A 2020 study conducted by Gallup and the NCAA examined undergraduate experiences and post-graduate outcomes for athletes and nonathletes. It reported that college athletes experienced better well-being outcomes than nonathletes, highlighting the potential long-term benefits of sports participation.

In summary, this analysis explored the relationship between sports participation and happiness using data from the World Values Survey. Findings offered limited support for a positive association as the results were constrained by the nature of the sports participation variable. Gender appeared to moderately affect the relationship, while age did not. Despite these limitations, this analysis highlights the importance of social connection, involvement in organizations, and physical activity in contributing to life satisfaction and well-being. Future research can provide more depth and explore other factors in this relationship.

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Appendix

The data used for this paper can be found here by selecting “USA 2017” and downloading this data file . The data for Figure 2 can be found here by using the World Values Survey online analysis to download these data files .

When creating Figure 2 (the scatterplot), the intention was to show correlation. Scatterplots are the most common data visualization to represent this (Schwabish, Ch.8, p. 249). The headline of the plot is informative and tells the reader the takeaway (Schwabish, Ch. 2, pp. 36-38). This plot has a simple and clear design, as the gridlines are removed and one color for the points, reducing the clutter of the visualization (Schwabish, Ch. 2, p. 31). Lastly, the vertical axis label is rotated horizontally to make it easier to read (Schwabish, Ch. 4, p. 79).

When creating Figures 3 through 8 (the side-by-side boxplots), using color was important to differentiate each variable. Starting with gray, allowed for intentional and meaningful color choices (Schwabish, Ch. 2, p. 44). Figure 3 was kept as gray because it was the foundational visualization. Figures 4 and 5 were blue and pink respectively to further signify gender as males are typically associated with blue and females are typically associated with pink (e.g. gender reveals). Figures 6, 7, and 8 were made to be different shades the same color to signify the continuous nature of the variable age. Orange was chosen as it stands out from the previous color choices. Unfortunately, Excel would not allow for the vertical axis labels to be rotated horizontally for easier reading, but that would have been implemented.

For the moderator variables, the visuals were placed side by side to allow for positional comparison, the most perceptually accurate encoding for readers (Schwabish, Ch. 1, pp. 13-14). Figures 6, 7, and 8 were not given axis titles as they followed the same format as previous boxplots. This allowed for the visualizations to be placed next to each other without being too small to comprehend.